

5.34 Solution: (a) From Problem 5.10 (b), the vector potential from a loop in the xy plane with current I is

$$A_\phi(\rho, z) = \frac{\mu_0 I a}{2} \int_0^{+\infty} e^{-k|z|} J_1(ka) J_1(k\rho) dk.$$

The interaction energy between this loop and another coaxial loop with the same current and a distance R away is simply

$$W_{12} = 2\pi a I A_\phi(a, R) = \mu_0 \pi a^2 I^2 \int_0^{+\infty} e^{-kR} J_1(ka)^2 dk.$$

Therefore,

$$M_{12} = \frac{W_{12}}{I^2} = \mu_0 \pi a^2 \int_0^{+\infty} e^{-kR} J_1(ka)^2 dk.$$

(b) Using the MacLaurin series of $J_1(x)$ and expand it to x^5 ,

$$J_1(x) = \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384},$$

the mutual inductance can be calculated as

$$\begin{aligned} M_{12} &= \mu_0 \pi a^2 \int_0^{+\infty} e^{-kR} \left(\frac{ka}{2} - \frac{(ka)^3}{16} + \frac{(ka)^5}{384} \right)^2 dk \\ &= \mu_0 \pi a^2 \int_0^{+\infty} e^{-kR} \left(\frac{k^2 a^2}{4} - \frac{k^4 a^4}{16} + \frac{5k^6 a^6}{768} \right) dk \\ &= \mu_0 \pi a^2 \left(\frac{a^2}{4} \frac{2}{R^3} - \frac{a^4}{16} \frac{24}{R^5} + \frac{5a^6}{768} \frac{720}{R^7} \right) \\ &= \mu_0 \pi a^2 \cdot \frac{1}{2a} \left[\left(\frac{a}{R} \right)^3 - 3 \left(\frac{a}{R} \right)^5 + \frac{75}{8} \left(\frac{a}{R} \right)^7 \right] \\ &= \frac{\mu_0 \pi a}{2} \left[\left(\frac{a}{R} \right)^3 - 3 \left(\frac{a}{R} \right)^5 + \frac{75}{8} \left(\frac{a}{R} \right)^7 \right]. \end{aligned}$$

(c) We have the result for M_{12} at $\theta = 0$. Since $\nabla_R^2 M_{12}(\mathbf{R}) = 0$, for arbitrary θ , we must have

$$M_{12}(R, \theta) = \frac{\mu_0 \pi a}{2} \left[\left(\frac{a}{R} \right)^3 P_2(\cos \theta) - 3 \left(\frac{a}{R} \right)^5 P_4(\cos \theta) + \frac{75}{8} \left(\frac{a}{R} \right)^7 P_6(\cos \theta) \right].$$

For the coplanar current loops, $\cos \theta = 0$, and

$$P_2(0) = -\frac{1}{2}, \quad P_4(0) = \frac{3}{8}, \quad P_6(0) = -\frac{5}{16},$$

and the mutual inductance becomes

$$M_{12} \left(R, \frac{\pi}{2} \right) = -\frac{\mu_0 \pi a}{4} \left[\left(\frac{a}{R} \right)^3 + \frac{9}{4} \left(\frac{a}{R} \right)^5 + \frac{375}{64} \left(\frac{a}{R} \right)^7 \right].$$

(d) The force between the current loops is given by

$$F = I^2 \nabla_R M_{12},$$

along the direction connecting the centers of the loops. For the coaxial case,

$$F = -\frac{3\mu_0\pi a^4 I^2}{2R^4},$$

and for the coplanar case,

$$F = \frac{3\mu_0\pi a^4 I^2}{4R^4},$$

to the leading order. We can see that these agree with Problem 5.18 (c), both of which have the form of magnetic dipole interaction. Here, the force for coplanar loops is repulsive, as the currents are in the same direction, while in Problem 5.18 (b), the current and its image current are in opposite directions.